Real Time Implementation on FPGA of an OFDM based Wireless LAN modem extended with Adaptive Loading

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Abstract

Future wireless applications target multimedia and high-speed internet access, all requiring techniques to improve the link capacity and robustness. For wireless Local Area Network (LAN) the standards have chosen orthogonal frequency division multiplex (OFDM) as modulation scheme. An extension beyond the standards to improve the link capacity is adaptive loading. This paper presents the implementation aspects of an OFDM based wireless LAN modem together with adaptive loading on field programmable gate array (FPGA). The FPGA implementation of the wireless LAN modem enhances rapid prototyping and allows flexible integration of extensions to improve the quality of service. Implementation results are given and a comparison is made with an ASIC implementation of the modem. The wireless LAN system will be demonstrated on a generic platform that is developed for prototyping and demonstration of high speed communication systems.

1. Introduction

The standards IEEE.802.11a and Hiperlan-2 have chosen for the upcoming broadband wireless LAN systems OFDM as modulation scheme because of its good performance for frequency selective channels in the 5 GHz band. The maximum data rate of 72 Mbit/sec in a 20 MHz bandwidth requires spectrally efficient coding schemes up to QAM-64. IMEC was one of the first to implement a broadband OFDM baseband modem [1] on an Application Specific Integrated Circuit (ASIC). This modem contains all functionality that is needed for OFDM burst transmission and for robust OFDM reception in a multi-path environment. The architecture of the ASIC is optimized for low power. The ASIC is highly programmable to adapt for different services and environments. In this paper we present the design aspects to map the OFDM modem on a FPGA.

The implementation on flexible hardware allows the addition of new functionality to the OFDM modem in order to improve the performance and the link capacity. One means to increase the capacity of wireless LAN is applying adaptive loading on top of the OFDM modulation. Simulations [2] show that a gain of 6 dB is achieved at BER of 10^{-2} and even better improvement is achieved at higher E_b/N_0. In the study done in [2] the Fischer et al [3] algorithm is selected because of its low computational complexity and its similar performance compared to other algorithms. This paper presents the implementation aspects of the adaptive loading algorithm and its integration with the OFDM baseband modem.

The OFDM modem is modeled in high level fixed point C++ dataflow and in VHDL register transfer. A simulation environment is set up with automatic comparison checks.

A generic demonstration platform is proposed on which the OFDM based wireless LAN modem is evaluated.

The next sections describe details of the modem architecture, the FPGA specific design aspects, the design flow and the implementation complexity comparison between ASIC and FPGA implementation. In the last section an overview of the demonstration platform is given.

2. Principles of OFDM and Adaptive Loading

The indoor propagation channel for wireless LAN applications is frequency selective due to multi-path fading. The link between transmitter and receiver is often a non-line of sight link. OFDM exploits the frequency diversity by partitioning the bandwidth in narrow sub-bands each seeing a flat fading channel. Each sub-band is PSK or QAM modulated.

Given the dips in the channel (see Figure 2) and OFDM as modulation scheme adaptive bit loading on the sub-carriers results in an improvement of the capacity usage of the channel. By measurement of the channel and by identification of the dips and peaks, an optimal bit loading over the sub-carriers can be calculated for transmission. The Fischer et al algorithm distributes the bits (R_t) over the different sub-carriers (D) in order to minimise the bit error rate (BER) at a constant total bit rate and transmit power. The bit assignment is given by:

$$R_i = \frac{R_t}{D} + \frac{1}{D}\log_2\left(\frac{\prod_{l=1}^{D} N_l}{N_i^D}\right) \qquad (1)$$

with N the equivalent channel power. The bit assignment has to be done recursively since R can become negative and those carriers have to be excluded.

3. Transceiver Architecture

The transceiver architecture (Figure 1) contains three main functional blocks:

1. OFDM (de)modulator
2. adaptive loading unit
3. debug and data analysis unit

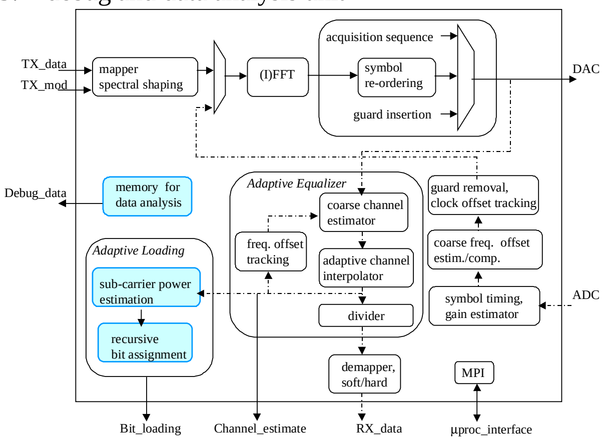

Figure 1 OFDM modem datapath architecture

3.1 OFDM (de)modulator

The architecture of the OFDM (de)modulator is derived from the architecture of its ASIC implementation. It is built of processes that operate stand-alone and that are activated by data tokens.

In the transmitter the data bits are mapped on N_c parallel signals with a programmable bit loading for each individual signal (0,1,2,4,6). It is then modulated on N_c parallel orthogonal carriers by means of an IFFT and a guard is added to create a OFDM symbol. For synchronisation purposes a preamble is inserted at the start of a burst.

The receiver performs basically the inverse of the transmitter and does synchronisation in time and frequency domain. The timing and the coarse frequency acquisition are done in a feed forward way before the FFT. The start of a burst is detected by power measurements and repeated correlation peak detection. The frequency offset is determined by correlation of two known sequences. The frequency offset compensation before the FFT reduces the inter-carrier interference. Clock offset tracking is performed by correlation on the guard. A simple equalisation can be implemented in frequency domain by a one tap equalizer per subcarrier. The equalizer performance is improved by adaptive interpolation to mitigate time variant channels and remaining frequency offset. The channel estimation is done on a known symbol.

Resource sharing between the transmitter and receiver chain is exploited for the (I)FFT and the data reordering tasks. A clock strategy for power optimisation is implemented by disabling the clocks of non active processes. This gives a significant average power saving for the receiver when it is listening to an incoming signal.

3.2 Adaptive Loading Unit

The adaptive loading unit at the receiver takes the channel estimate of the equalizer as input to calculate the equivalent channel power on the sub-carriers and to recursively calculate the bit loading until the target bit rate is obtained. Figure 2 shows the channel profile, the estimated channel profile by the adaptive equalizer and the optimal bit loading calculated by the adaptive loading unit.

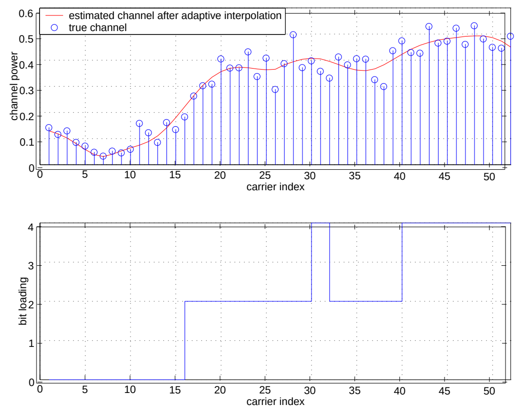

Figure 2 Indoor frequency channel response and corresponding bit loading

The execution time of the algorithm should be minimised and should be at least less than two OFDM symbols (8 µs). The inputs for the algorithm are generated serially at a clock rate of 20 MHz. Therefore, the first part of the algorithm, calculating the channel power and the logarithm, is also done serially at 20 MHz. The remainder of the algorithm, i.e. the recursive bit loading, is implemented fully parallel. The maximum execution time of the adaptive loading implementation is determined by simulating the bit loading for 10000 Hiperlan-2 channels. This results in a maximum execution time of 4.25 µs for the recursive bit loading implementation clocked at 40 MHz and in a maximum execution time of 5.75 µs for the recursive bit loading implementation clocked at 20 MHz.

The fixed point implementation gives an extra signal to noise loss in the BER performance curve. The BER performance curve, done on 10000 Hiperlan-2 channels, is shown in Figure 3 for the register transfer VHDL implementation. The signal to noise implementation loss is 0.09 dB at the bit error rate of 10^{-5} compared with the floating point Fischer algorithm.

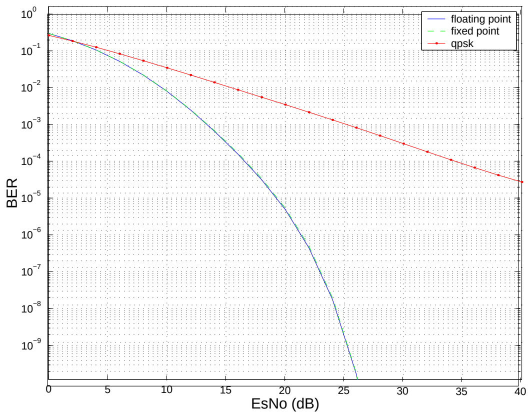

Figure 3: BER comparison for the fixed point register transfer and the floating point Fischer algorithm

3.3 Debug and Data Analysis Unit

Memory is allocated to store debug data and channel measurements. This data can be processed and analysed off line. The following data can be stored in memory:

- a OFDM burst of 2 ms at 20 MHz sampling rate at the transmitter output (DAC) and receiver input (ADC)
- channel estimations for channel profiling
- internal signals of the OFDM modem for debug purposes

4. Adaptations for FPGA implementation

The register transfer VHDL netlist of the OFDM modem, that is optimised for low power consumption and ASIC implementation, is used as basic netlist for mapping on a XilinX Virtex-2 FPGA [4]. The adaptations required for FPGA implementation are related to the clock distribution network and the timing constraint of 50 ns.

The FPGA Digital Clock Manager (DCM) with PLL is used to derive the main internal clock at 20 MHz out of the external input clock at 40 MHz. The DCM installs a zero phase delay between the internal and external clock and this allows to use a FIFO interface operating on the clock edges to transfer transmit data (TX_data) and received data (RX_data). The ASIC implementation is optimised for power consumption by disabling the main internal clock when a functional unit is not operational. This derived clock is the output signal of a buffered AND gate with the main internal clock and the control enable signal as input signals. In inactive mode the derived clock is low. On FPGA the derived clocks are implemented with a BUFGCE clock buffer that has the same behaviour as the buffered AND gate. In this way a power optimised implementation is realised of the OFDM modem on the FPGA.

The mapping on the FPGA did give timing violations for some functional units. This was the case in the equalizer where additional pipeline registers are added in the divider and in the datapath of the channel estimator and interpolator.

5. Design Flow

The OFDM (de)modulator is described as a high level dataflow model in C++ using the OCAPI [5] hardware libraries. Algorithmic exploration, performance simulations and fixed point refinement are done on this model. The C++ dataflow model is further refined to a C++ description of combined finite state machine and datapath (FSMD). From this description the VHDL code is automatically generated. The channel is modelled in C++ dataflow with programmable frequency offset, timing offset, noise insertion, multi-path channel and up- and down-sampling filter.

The VHDL register transfer model of the adaptive loading unit is manually written. The interface between the OFDM (de)modulator and the adaptive loading unit is written in VHDL.

A simulation environment is defined to test the OFDM modem whereby an automatic comparison check is done on the internal signals and on the output signals between the fixed point C++ dataflow model and the VHDL netlist.

6. Implementation Results

The XilinX Virtex2 family is selected as target FPGA for the mapping of the OFDM modem extended with adaptive loading. The design did not fit into the XC2V3000 FPGA because the number of available multipliers is lower than the required number and this leads to an inefficient implementation of the remaining multipliers on the slices. The implementation figures for FPGA are summarised in **Table 1** and for ASIC implementation in **Table 2**. The FPGA implementation has as extra functionality the adaptive loading unit and the memory (1.25 Mbit) for off line data analysis. Remark that the equalizer takes besides 36.2 % of the slices also 79% of the multipliers.

Table 1. FPGA key figures of OFDM modem with adaptive loading

FPGA	XC2V6000
Internal main clock	20 MHz
Slices	16167 = 47,8 % usage
- active in reception	15309 = 94.7 %
- active in transmission	2650 = 16.4 %
- of which equalizer	5862 = 36,2 %
- of which fft	1963 = 12,1%
- of which ad. loading	4103 = 25,4%
Multipliers	124 = 86% usage
- of which in equalizer	98 = 79%
- of which in fft	8 = 6.4%
Block RAM	81 = 56.2 %
- of which functional	8 = 9.8%
- of which data analysis	73 = 90 %

Table 2: ASIC key figures of OFDM modem

Technology	CMOS 0.18 µm 1.8 V core, 3.3 V I/O
clock	20 MHz
Equ. Gate count (core)	431000 = 100.0 %
- active in reception	416000 = 96.7 %
- active in transmission	79000 = 18.4 %
- of which equalizer	270000 = 62.6 %
- of which fft	42000 = 9.7%
- of which RAMs	78000 = 18.1%
Die Size	20.8 mm^2

7. Demonstration Platform

We have defined generic platform concepts [6] to enable reuse of modular hardware and of Linux driver development software. The hardware concepts feature dedicated high-speed inter-board data links, flexible configurable hardware, integration of Intellectual Property (IP) cores and built-in debug facilities. The boards are Compact PCI compliant and can be plugged in a standard shelf to build a system. The wireless LAN system will be demonstrated on this platform (see Figure 4) for which two boards are developed. One board is a general purpose board with two XilinX Virtex2 family FPGAs for implementation of the application specific cores. The OFDM modem with adaptive loading is mapped on a XC2V6000 FPGA. The second board contains one XilinX Virtex2 family FPGA and a socket to mount a front-end daughter board. The automatic gain control (AGC) and intermediate frequency up- and down conversion is implemented on the FPGA. For the real time demonstration of the wireless LAN system, the communication of payload data between the boards is managed via high speed data links with a capacity of 1.4 Gbit/sec per link.

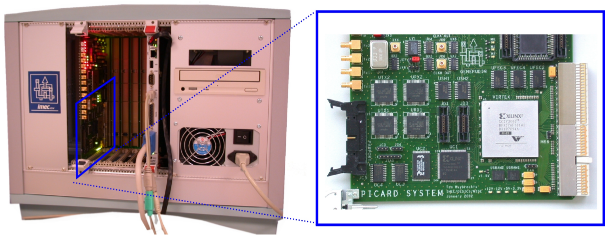

Figure 4: Demonstration Platform for high speed telecommunication systems, e.g wireless LAN system

8. Conclusions

In this paper, the implementation on FPGA of a broadband OFDM modem, which achieves data rates up to 72 Mbit/sec, is presented. It contains all functionality for (de)modulation, e.g. synchronisation and frequency domain equalisation, and also contains adaptive loading to improve the capacity usage. Imec has developped an OFDM (de)modulator ASIC. The register transfer netlist of the ASIC, which is optimised for power consumption, is used as basic netlist for FPGA mapping. Modifications needed to be done on the clock distribution network to disable clocks of non active processes for power optimisation. Extra pipeline registers are added in the FPGA implementation to meet the timing constraints. The implementation takes 48% of the slices and 86% of the multipliers in a XC2V6000 FPGA. The adaptive loading takes 25% of the slices of the OFDM modem and gives a performance improvement of 6 dB at BER of 10^{-2}

A generic platform for high speed systems is proposed on which the wireless LAN system will be demonstrated.

8. References

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